



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-Section SYSTEMS INTEGRATION AND ARCHITECTURE 1

FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

Project Title: Mini App – User Registration & Authentication

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Date of Submission: February 8, 2026

Version: 1

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1. Introduction

1.1. Purpose

The purpose of this document is to describe the design and system flow of a user authentication system, including user registration, login, logout, and access to protected pages. It is intended to provide students, instructors, and developers with a clear understanding of the system behavior through diagrams and documentation.

1.2. Scope

The system functionality will:

- Allow users to register an account
- Allow users to log in and log out
- Allow authenticated users to access their profile or dashboard
- Restrict access to protected pages for unauthenticated users

This system focuses only on basic user authentication and access control. Features such as advanced security, user roles, and account recovery are outside the scope of this design.

1.3. Definitions, Acronyms, and Abbreviations

- Authentication – The process of verifying a user's identity using login credentials.
- Authorization – The process of controlling access to protected system features.
- Guest User – A user who has not logged in and has limited access to the system.
- Authenticated User – A user who has successfully logged in and can access protected pages.
- ERD (Entity Relationship Diagram) – A diagram that shows the structure of the database.
- JWT (JSON Web Token) – A token used to maintain user login sessions.
- API (Application Programming Interface) – A set of endpoints that allows communication between the frontend and backend.

2. Overall Description

2.1. System Perspective

The system is a standalone authentication module integrated with a web or mobile app. The frontend communicates with the backend via APIs, which interact with the database to manage users and sessions using JWT tokens. Only authenticated users can access protected pages, while guests are limited to registration and login.

2.2. User Classes and Characteristics

The system has two types of users: **Guest Users**, who can register and log in but cannot access protected pages, and **Authenticated Users**, who have successfully logged in, can access their profile or dashboard, and can log out.

2.3. Operating Environment

Hardware: Any internet-enabled device (PC, laptop, or mobile).

Software: Web browser; Windows, macOS, Linux, Android, or iOS.

Tools: Spring Boot (backend), React (frontend), MySQL (database), Draw.io(diagrams).

2.4. Assumptions and Dependencies

- Users provide valid, unique email addresses during registration.
- The system relies on a backend database (MySQL) for storing user data.
- The frontend communicates with the backend via APIs.
- Sessions are managed using JWT tokens.

3. System Features and Functional Requirements

3.1. Feature 1: User Registration

Description: Allows a new user to create an account.

Functional Requirements:

- Fill in registration form.
- Validate input and check for existing email.
- Create account and display success/error message.

3.2. Feature 2: User Login

Description: Allows a registered user to access protected pages.

Functional Requirements:

- User enter email and password.
- System checks if the email exists and password is correct.
- Grant access or display error message.

3.3. Feature 3: User Logout

Description: Allows an authenticated user to end their session.

Functional Requirements:

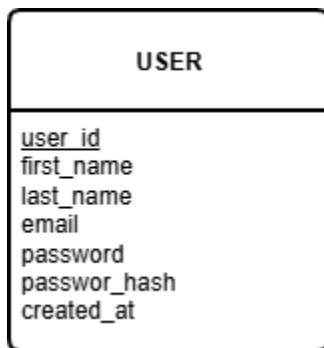
- System ends the user's login session so they can no longer access protected pages.
- Prevent access to protected pages until the user logs in again.

4. Non-Functional Requirements

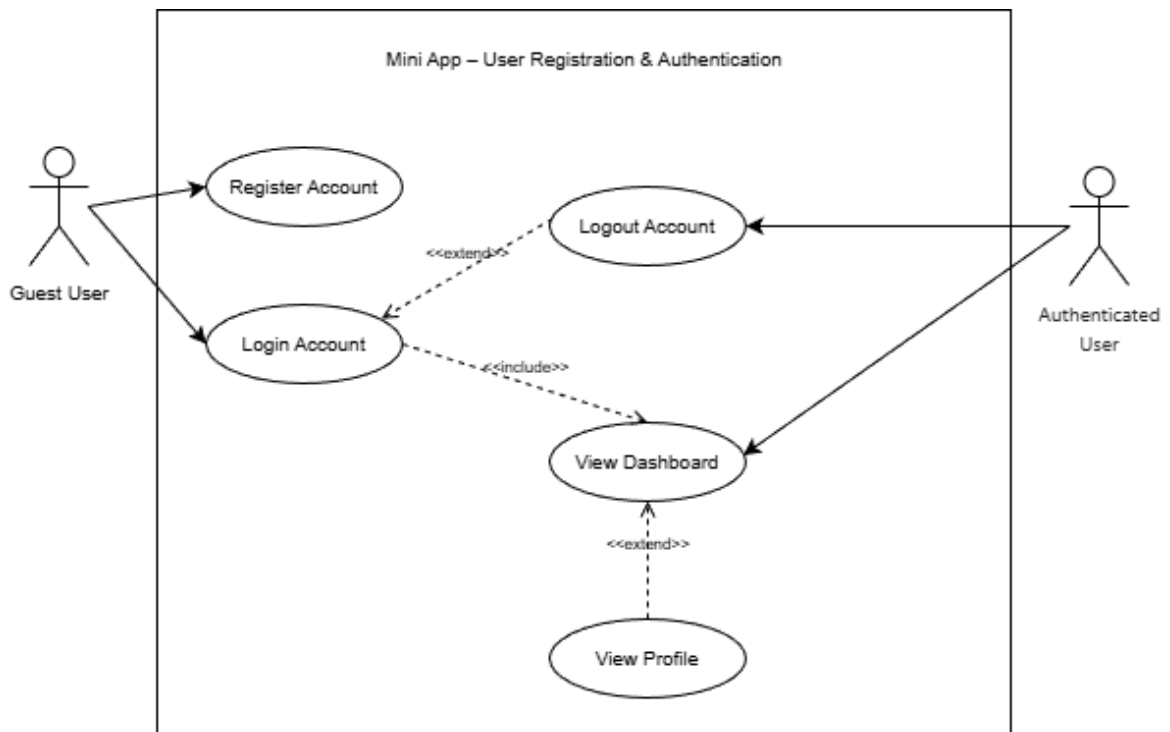
- Performance: The system should respond to actions like login, registration, logout, and viewing the dashboard within a few seconds.
- Security: Passwords must be stored securely (hashed), and users cannot access protected pages without logging in.
- Usability: The interface should be simple and easy to use for both web and mobile users.
- Reliability: The system should handle errors gracefully, such as incorrect login credentials or duplicate registration.
- Compatibility: The system should work on standard browsers and devices typically used by students.

5. System Models (Diagrams)

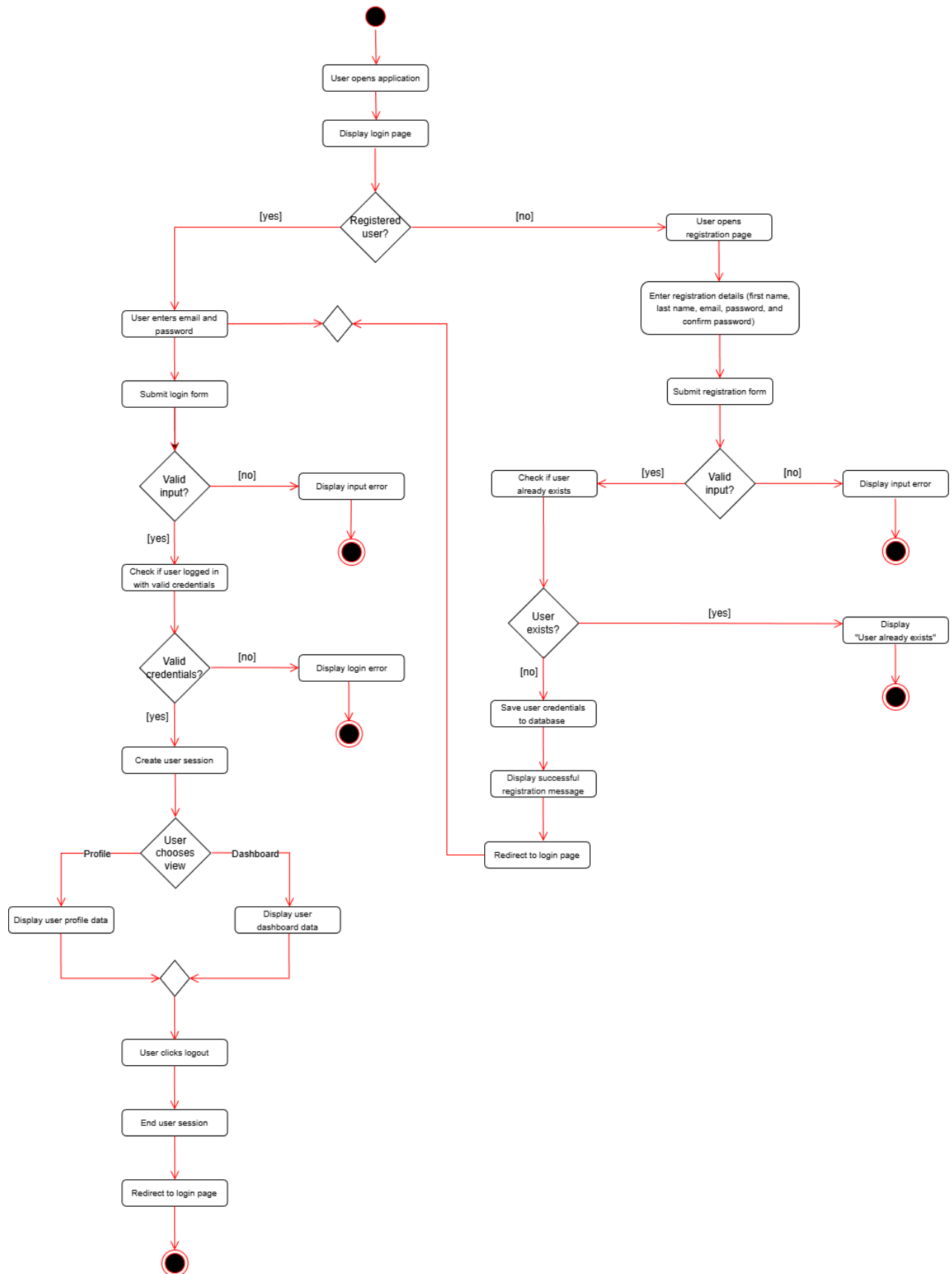
5.1. ERD



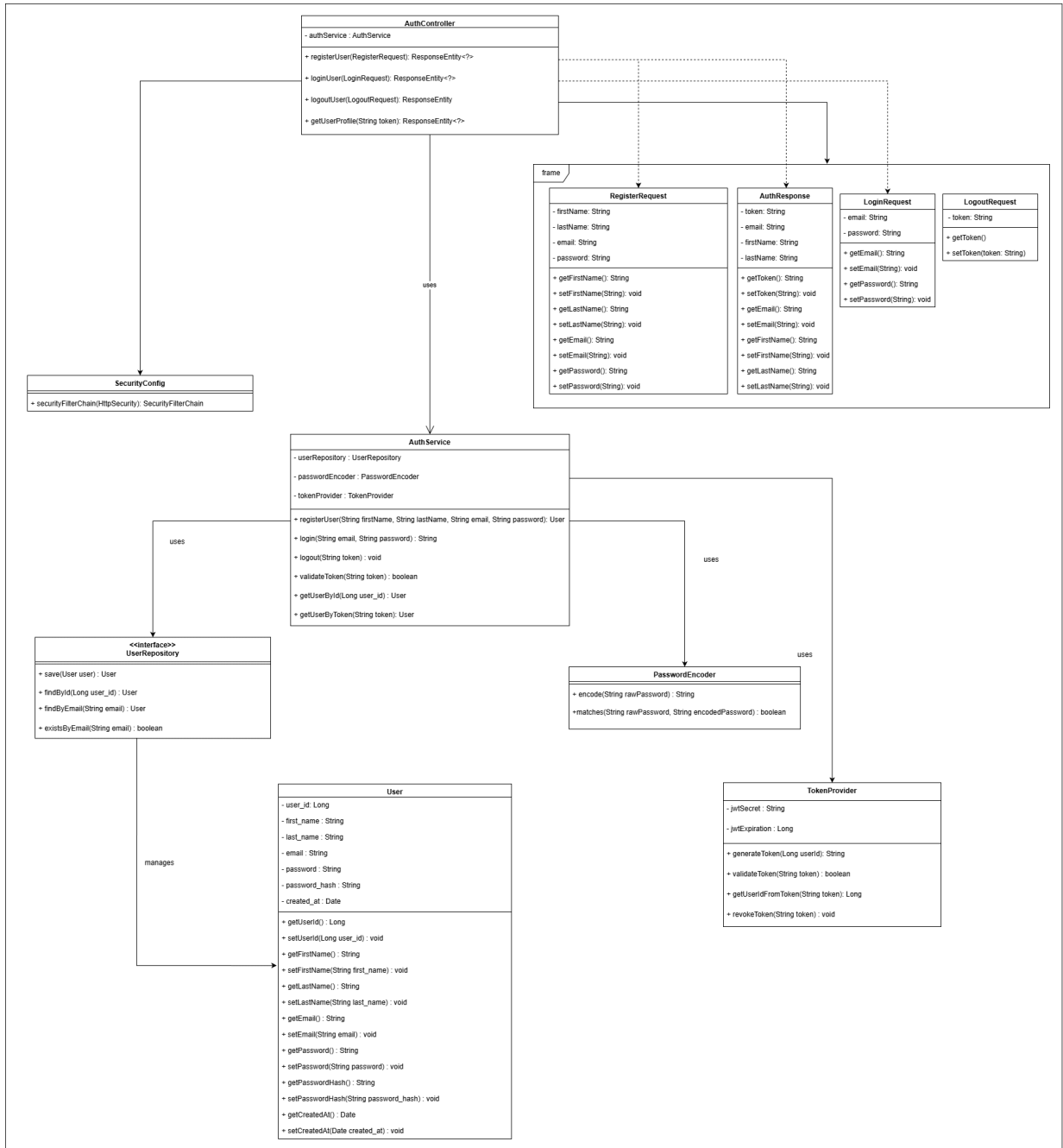
5.2. Use Case Diagram



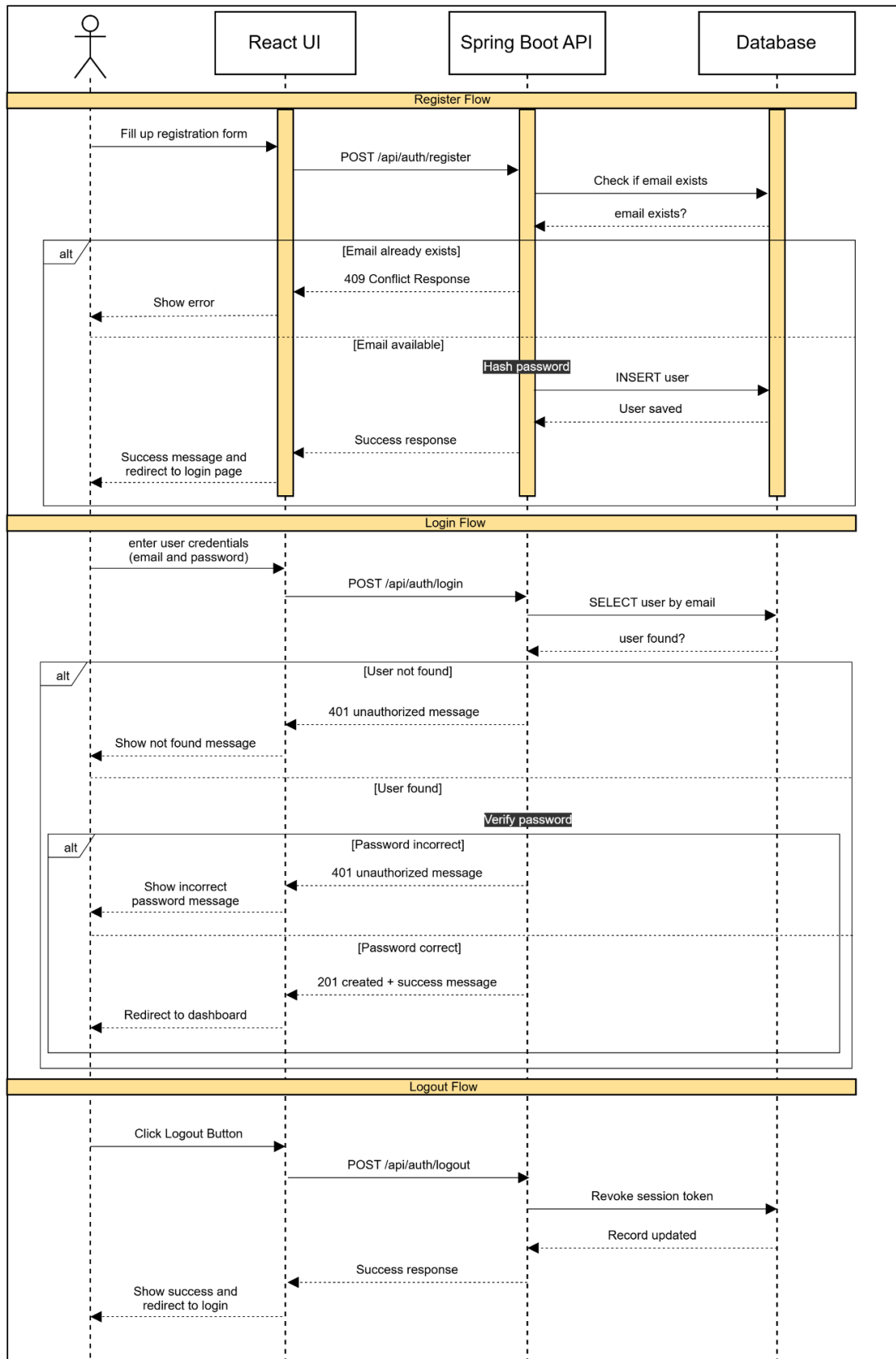
5.3. Activity Diagram



5.4. Class Diagram

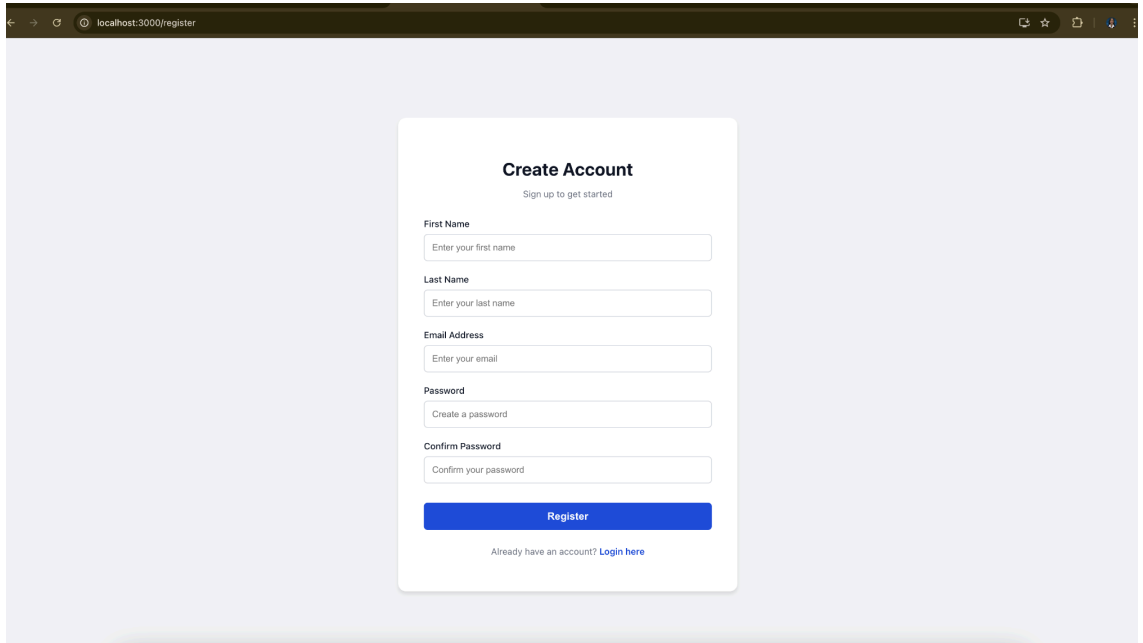


5.5. Sequence Diagram



Screenshots of the Web UI:

- Register



A screenshot of a web browser showing the 'Create Account' registration form. The browser's address bar displays 'localhost:3000/register'. The form is centered on a light gray background. It features a title 'Create Account' with a subtitle 'Sign up to get started'. Below the title are five input fields: 'First Name' (placeholder: 'Enter your first name'), 'Last Name' (placeholder: 'Enter your last name'), 'Email Address' (placeholder: 'Enter your email'), 'Password' (placeholder: 'Create a password'), and 'Confirm Password' (placeholder: 'Confirm your password'). A blue 'Register' button is positioned below the inputs. At the bottom, a link 'Login here' is provided for users who already have an account.

localhost:3000/register

Create Account

Sign up to get started

First Name
Enter your first name

Last Name
Enter your last name

Email Address
Enter your email

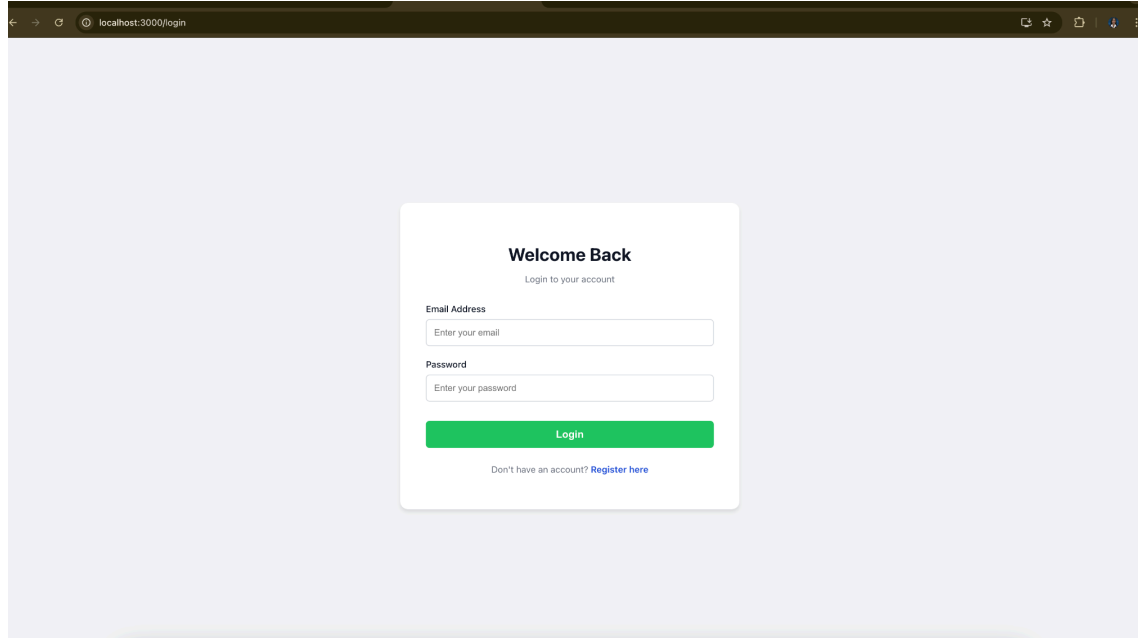
Password
Create a password

Confirm Password
Confirm your password

Register

Already have an account? [Login here](#)

- Login



A screenshot of a web browser showing the 'Welcome Back' login form. The browser's address bar displays 'localhost:3000/login'. The form is centered on a light gray background. It features a title 'Welcome Back' with a subtitle 'Login to your account'. Below the title are two input fields: 'Email Address' (placeholder: 'Enter your email') and 'Password' (placeholder: 'Enter your password'). A green 'Login' button is positioned below the inputs. At the bottom, a link 'Register here' is provided for users who don't have an account.

localhost:3000/login

Welcome Back

Login to your account

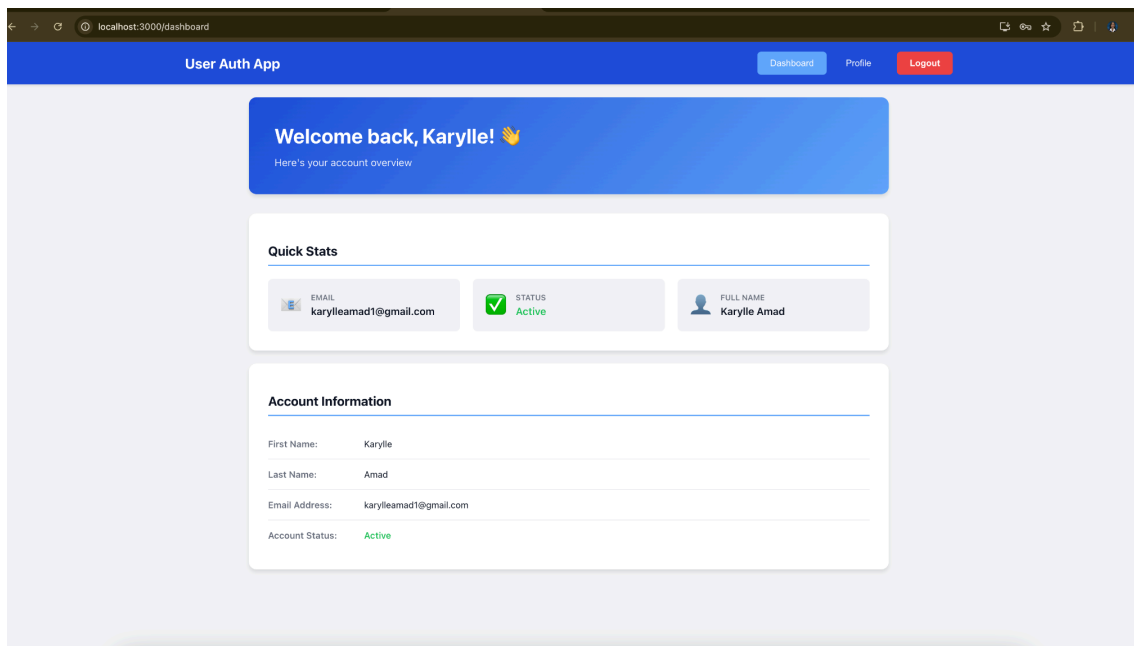
Email Address
Enter your email

Password
Enter your password

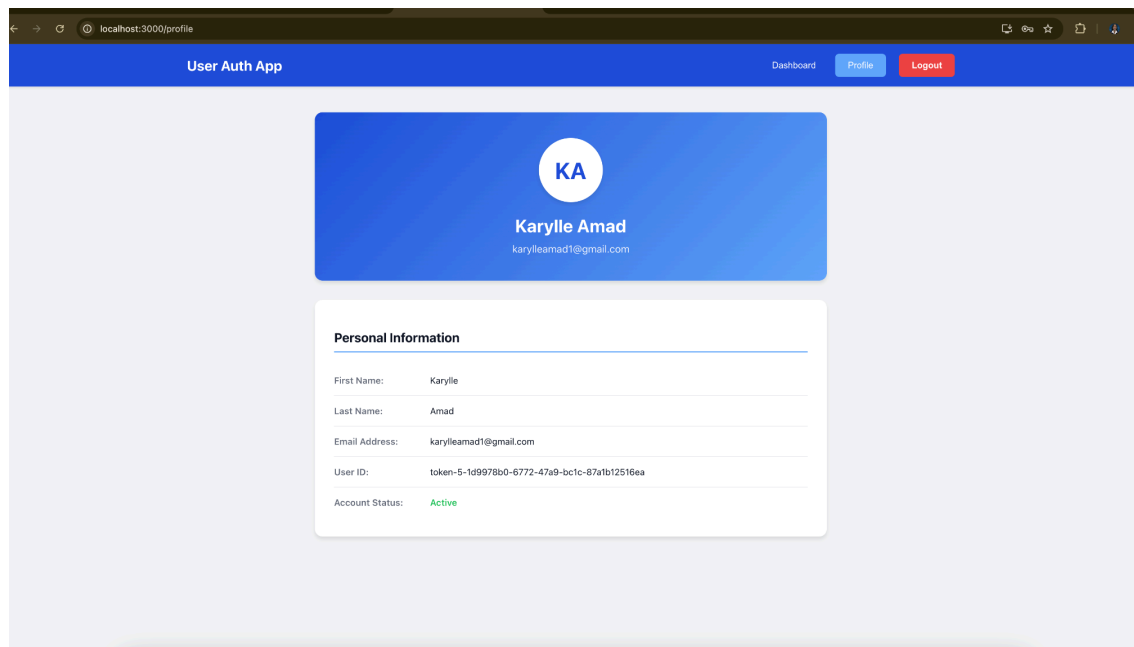
Login

Don't have an account? [Register here](#)

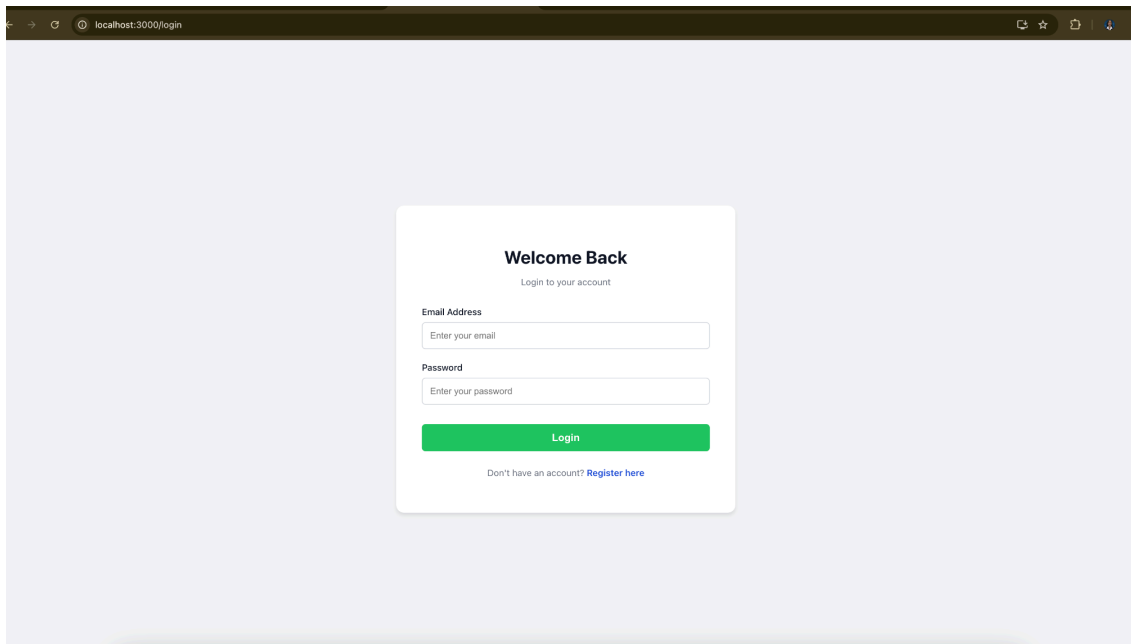
- Dashboard



- Profile

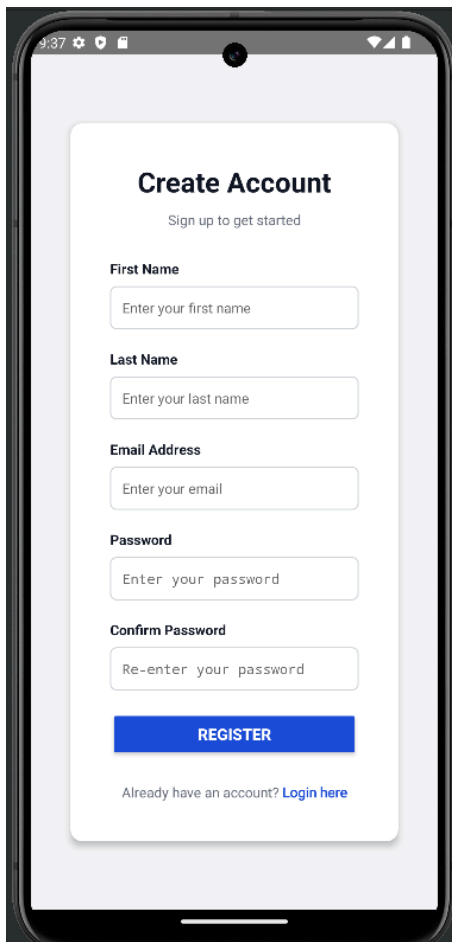


- Logout

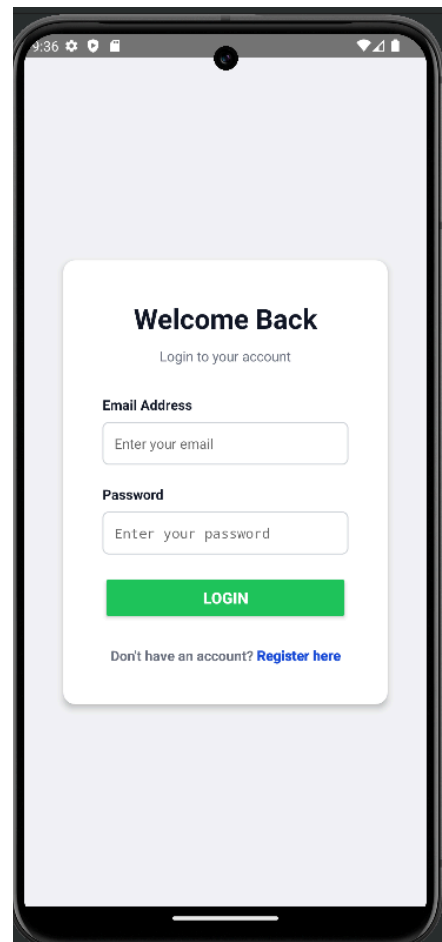


Screenshots of the Mobile Application UI:

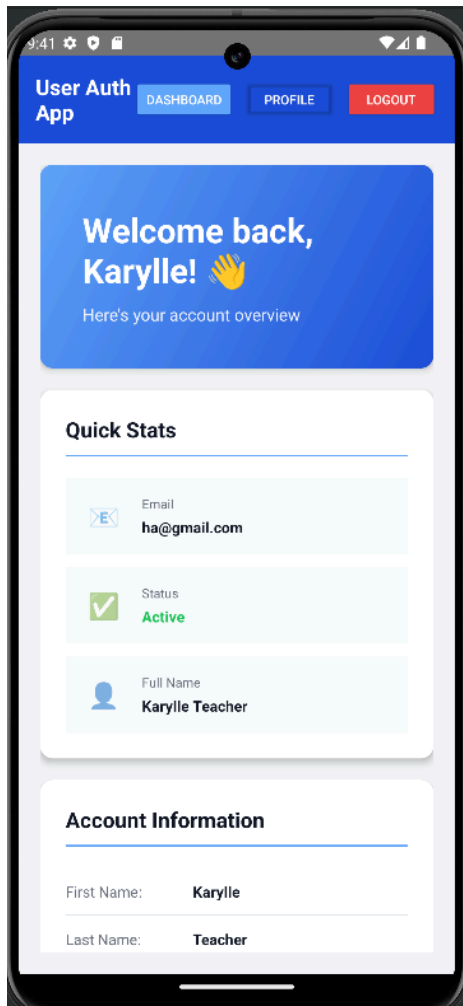
- Register



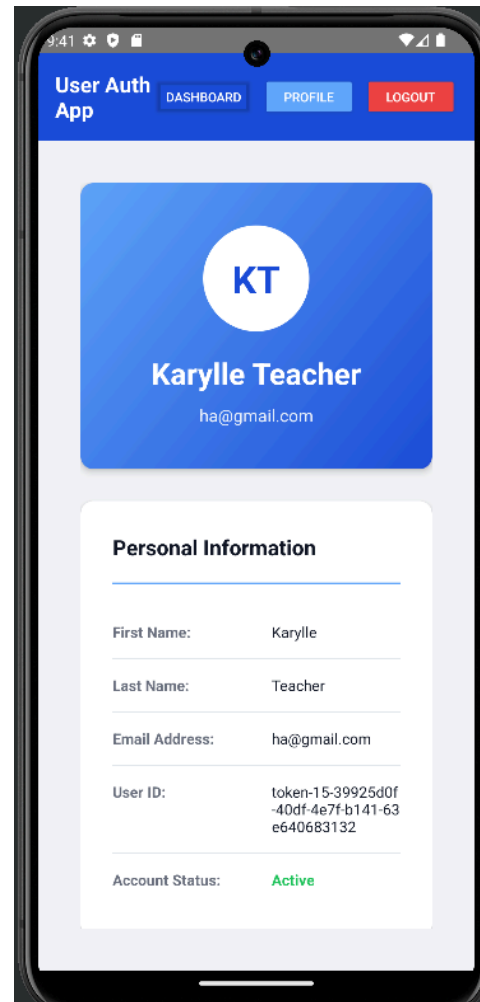
- Login



- Dashboard



- Profile



6. Appendices

References:

Visual Paradigm. What is a Class Diagram?

<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-class-diagram/>

Visual Paradigm. What is a Sequence Diagram?

<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-sequence-diagram/>

Visual Paradigm. What is an Activity Diagram?

<https://www.youtube.com/watch?v=rsTWufuP328>

<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-activity-diagram/>

Support Materials:

- Sample diagrams (ERD, Use Case, Activity, Class, Sequence)
- Mockups or screenshots of the user interface used for design reference